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Geomagnetic Jerks: A Study Using Complex Wavelets

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Introduction

The naturally produced geomagnetic field variations recorded at the earth's surface or at satellite altitudes is of dual origin: external and internal. The external part originates from the current systems generated in the ionosphere and the distant magnetosphere by the interaction of the solar wind with them. As a result, different types of geomagnetic variations based on the solar intensity having different periodicities are generated. A small portion of the internal part arises due to the eddy currents induced in the conductive layers of the earth by the time-varying external field diffusing within the earth. In addition to induction by external sources, the major contribution for the internal part, known as the earth's permanent magnetic field or dipole field, originates from the earth's liquid outer core. This is a slowly varying field with time. More than 95% of the earth's magnetic field is believed to be due to the electric currents generated within the fluid outer core. An additional contribution of approximately 2% comes from the earth's lithosphere and crust. Thus, the science of geomagnetism opens up a plethora of research topics related to terrestrial and extraterrestrial phenomena, and facilitates the studies of external source current systems, secular variations, and electrical conductivity variations within the earth. Figure 7.1 depicts a schematic of the broad classification of geomagnetic field variations. Figure 7.2 describes the vector representation of the earth's magnetic field at any point in the northern hemisphere of the earth.

Figure 7.3a shows an example of long-period geomagnetic field variations, whose first time-derivative, known as geomagnetic secular variations (Figure 7.3b), owe their origin to the convection currents generated by the dynamo processes actively taking place within the earth's liquid outer core. It has been observed that the geomagnetic data of several decades show some sudden changes in the slope of the secular variations (i.e., second time-derivative of the long-period geomagnetic field), having periods of approximately 1 year. Such features are generally known as secular accelerations or geomagnetic jerks (GJs; Figure 7.3c; Courtillot et al. 1978; Courtillot and Le Mouel 1976, 1984; Backus 1983; Gubbins and Tomlinson 1986; Mandea et al. 2000). Barraclough et al. (1989) described the use of horizontal components of the magnetic field to understand the outer core motions. GJs are easily noticeable in the east–west component (*Y* or *D*, Figure 7.2) of the magnetic field, although it is sometimes seen in the north–south component (*X* or *H*, Figure 7.2) as well (Bloxham et al. 2002). Well-known global jerks have occurred

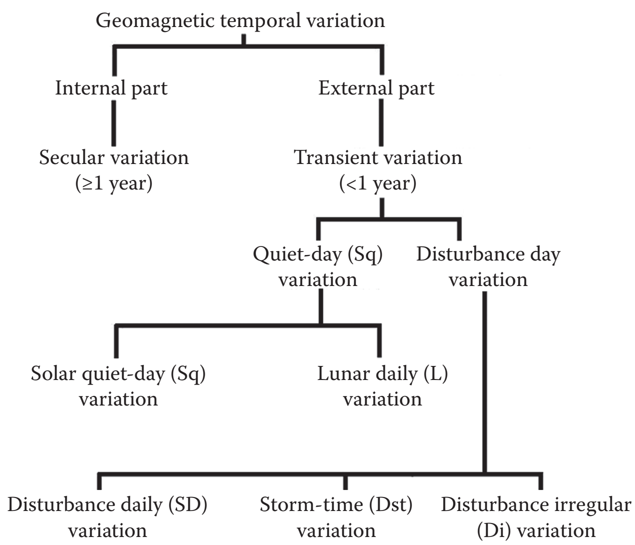

FIGURE 7.1
A schematic showing different types of temporal geomagnetic field variations.

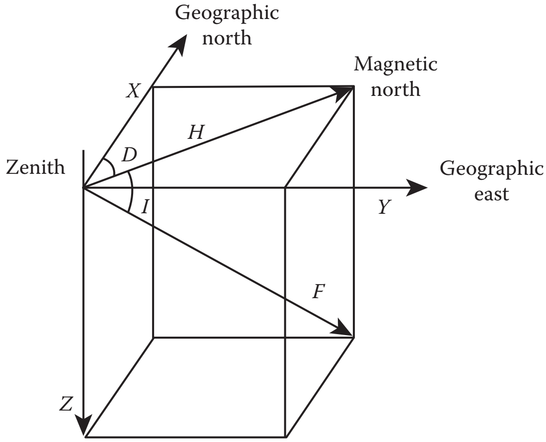

FIGURE 7.2
Elements of the geomagnetic field in northern hemisphere. The declination (D) refers to the angle between geographic north (X) and geomagnetic north (H). The inclination (I) refers to the angle between the geomagnetic north and the total field (F). Y denotes the geographic east component and Z refers to the direction of the vertical magnetic field.

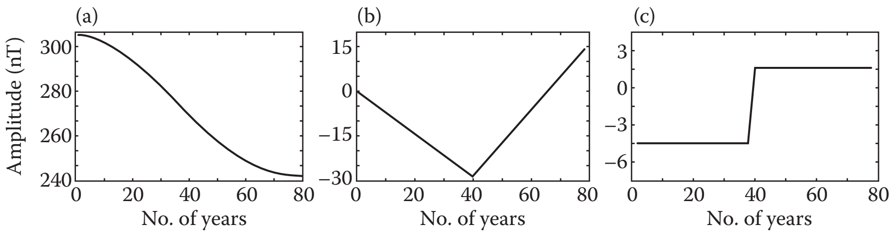

FIGURE 7.3
A schematic for explanation of GJ phenomenon. (a) Example of decadal variations of geomagnetic east–west component, Y; (b) secular variations, obtained by taking dY/dt; (c) secular acceleration (d^2Y/dt^2) or GJ, showing a sudden jump at the point of inflection in (a).

during the years 1969 to 1970, 1978 to 1979, and 1991 to 1992 and local jerks during 1901 to 1902, 1913 to 1914, 1925 to 1926, 1932 to 1933, 1942 to 1943, 1949 to 1950, around 1985, and in 1999 to 2000 (Alexandrescu et al. 1995, 1996; De Michelis et al. 1998; De Michelis and Tozzi 2005; Adhikari et al. 2009).

The origin of GJs has been the subject of a lot of debate. Employing a spherical harmonic analysis (SHA) technique, Malin and Hodder (1982) reported that GJs are of internal origin. Later, claiming some loopholes in Malin and Hodder's analysis, Alldredge (1984) argued that GJs are of external origin. However, further studies by Gavoret et al. (1986), Davis and Whaler (1997), Le Huy et al. (1998), Nagao et al. (2002), and Bloxham et al. (2002) confirmed the internal origin of GJs. Gavoret et al. (1986) separated the internal and external field components using geomagnetic indices and found a negligible